FINAL-YEAR RESEARCH PROTOTYPE · DESIGN SCIENCE RESEARCH

WiseLanka

Policy-aware education and career guidance.
Academic evidence first. AI relevance kept separate.

Author / student number / supervisor: to be confirmed · No university logo or template assumed

REVIEW DRAFT
University of West London
Official poster format pending
Evidence: 29 October 2026

01 · Problem & aim

How can an AI-powered, policy-aware Education and Career Pathway Intelligence System support Sri Lankan learners through personalised, explainable, and accessible pathway recommendations?

Design, develop and evaluate personalised, explainable and accessible AI-powered, policy-aware guidance for Sri Lankan learners.

02 · Curated educational knowledge

11
programmes

21
source records

IT:7 · Business:1 · Engineering:3
Five conditional/non-guaranteed progression relationships.

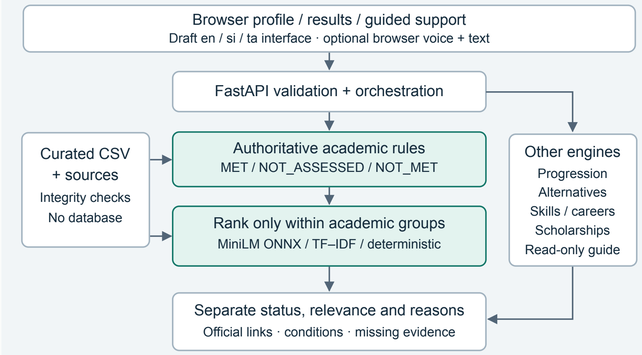
CSV records retain source links, conditions and verification dates. Reported qualifications do not establish equivalence.

03 · Research method

DSR with iterative prototyping: construct → demonstrate → evaluate → refine.

Source review found omitted UGC conditions. Corrections retain **NOT_ASSESSED** until unresolved evidence is verified.

04 · Hybrid architecture



Similarity never overrides eligibility. Guidance is not admission.

One modular FastAPI application; local inference, curated CSV knowledge and an existing web interface.

05 · Relevance without policy override

1. Rules establish academic assessment groups.
2. Frozen MiniLM ONNX ranks within groups.
3. TF–IDF and deterministic orders provide comparisons.
4. Facts, scores and official links remain distinct.

> Similarity is not admission probability.
> No automatic feedback learning.

06 · Measured technical evidence

205
backend tests passed

36
browser tests passed

Five live workflows ·14 synthetic reference cases passed ·CSV integrity and fresh installation passed.

Nine synthetic ranker comparisons; independent relevance labels unavailable.

0.29s model load ·**15ms** warm in-process p95 ·**302MiB** whole-process peak RSS.

Single Mac, sequential local requests; not production throughput or isolated model memory.

07 · Access & interaction

Guided support ·draft en/si/ta controls ·optional browser voice with text fallback.

Zero automated violations in nine scanned states. This is not WCAG conformance certification.

08 · Limits & next evaluation

Small uneven catalogue. Adult usability, independent relevance/source review, fluent translation and native speech/manual accessibility checks remain pending.

No learner-outcome, satisfaction or ranking-quality improvement is claimed.

Selected foundations: Hevner et al. (2004) ·Reimers & Gurevych (2019), DOI10.18653/v1/D19-1410 ·UGC Handbook2025/2026.

Evidence checkpoint:07f4b97
Full sources, logs and limitations accompany the draft.